INVENTORY DEMAND FORECASTING

Kalman Filtering + Global Gradient Boosting

PHASE 1 • M5 FORECASTING ACCURACY DATASET



Team project presentation

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Why forecast retail demand?

Retail demand changes across products, stores, weekdays, and events. Better forecasts can support replenishment planning and reduce avoidable uncertainty.

Forecast target

Daily unit sales for each item store time series.

Core challenge

Demand is intermittent and affected by weekly patterns, events, and store differences.

Project goal

Compare simple baselines and learnable models under the same fair evaluation.

Research question

Can a global model improve daily demand forecasts over a seasonal baseline, while a Kalman filter provides a useful uncertainty estimate?

DATA

Dataset & experiment scope

30,490

Item store sales series in the original sales table

1,969

Calendar records

6.84M

Sell price records

100

Sampled series: 10 from each of 10 stores

Input files

- sales_train_validation.csv : daily unit sales
- calendar.csv : dates, weekdays, events, SNAP indicators
- sell_prices.csv : item/store/week price information

Evaluation slice

- 100 product store series
- Four rolling origins
- Seven forecast days per origin
- 2,800 forecast points per model

End-to-end workflow

01

Explore

Data quality, sales patterns, weekday effects

02

Engineer

Calendar features, lags, rolling averages

03

Train

Seasonal Naive, Kalman, Gradient Boosting

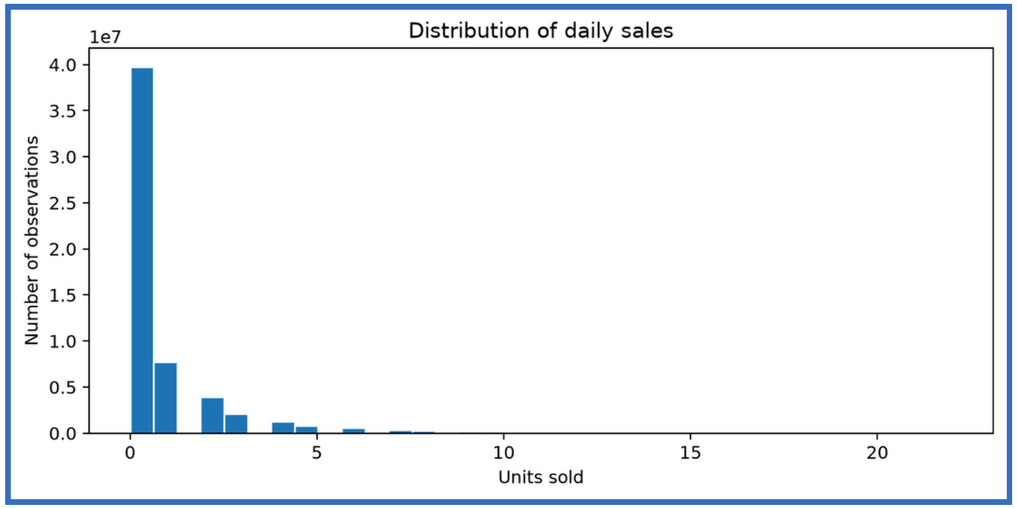
04

Evaluate

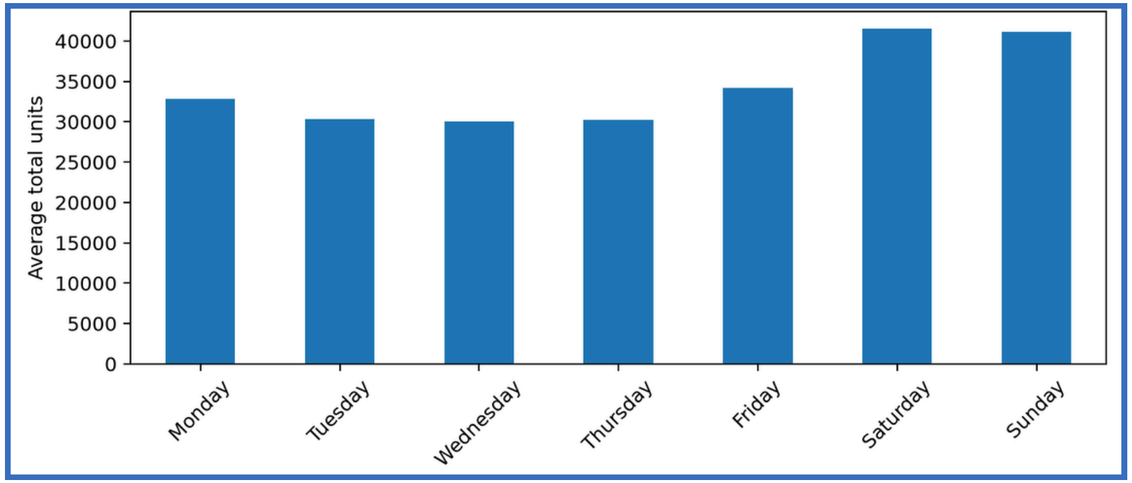
Four rolling windows; MAE, RMSE, WAPE

All models use the same sampled series, forecast origins, dates, and actual values.

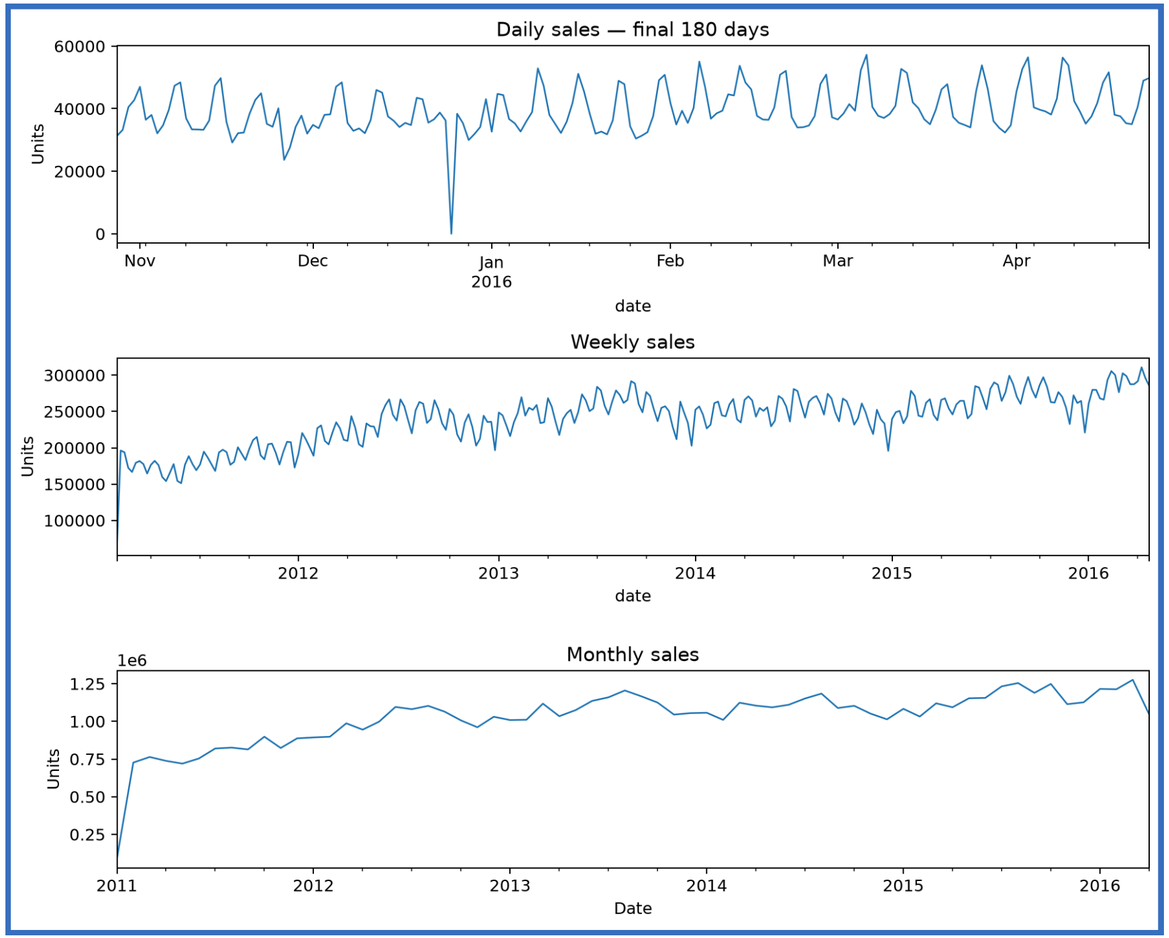
What does the demand look like?



Demand distribution



Average sales by weekday



Aggregate sales over time

Feature engineering

Features are built only from information available at the forecast origin.

Demand history

Lag 1, lag 7, lag 28

Recent level

7-day and 28-day rolling means

Context

Weekday, events, SNAP, item and store IDs

Leakage prevention

- Lag and rolling features are shifted to use past demand only.
- Training rows stop at each rolling forecast origin.
- Within each 7-day forecast, predictions - not hidden actual sales - feed later steps.

Three models, complementary roles

1. Seasonal Naive

Repeats the demand observed on the matching weekday one week earlier.

Role: transparent baseline.

2. Local-level Kalman

Updates a latent demand level as each new observation arrives.

Role: smoothing and approximate uncertainty intervals.

3. Global Gradient Boosting

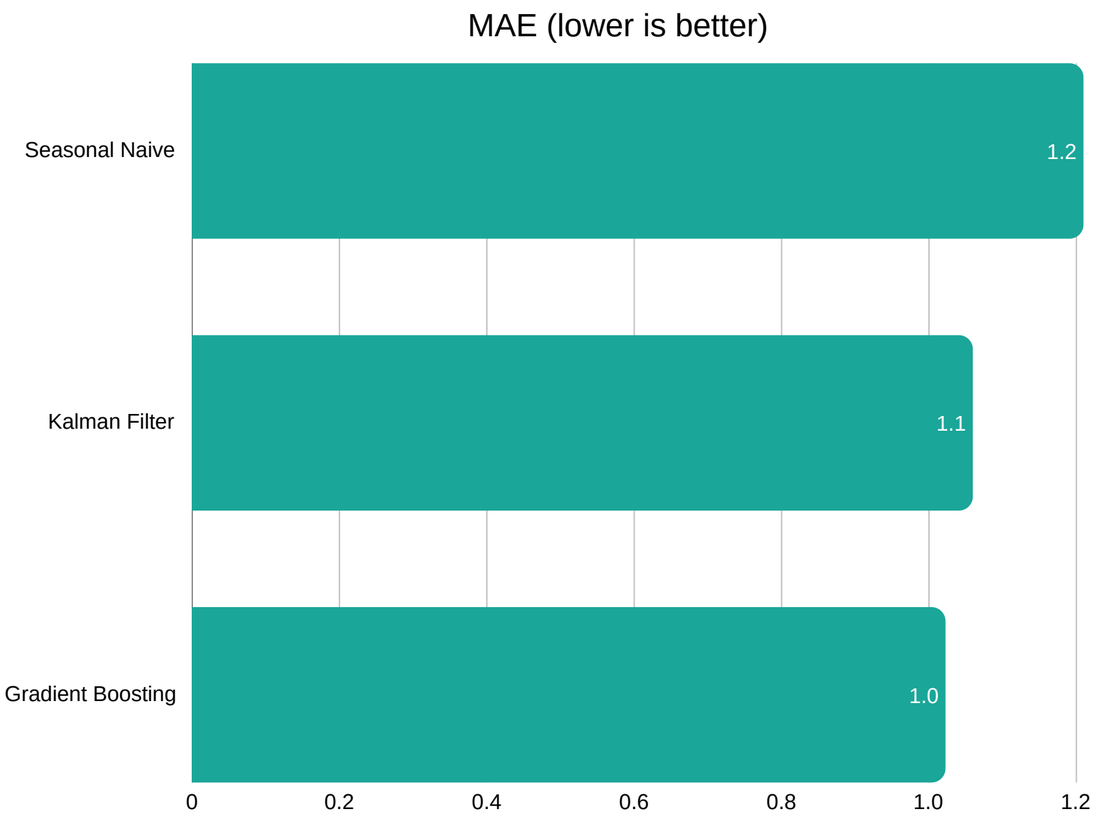
Learns shared patterns across sampled item–store series using lag, rolling, and calendar features.

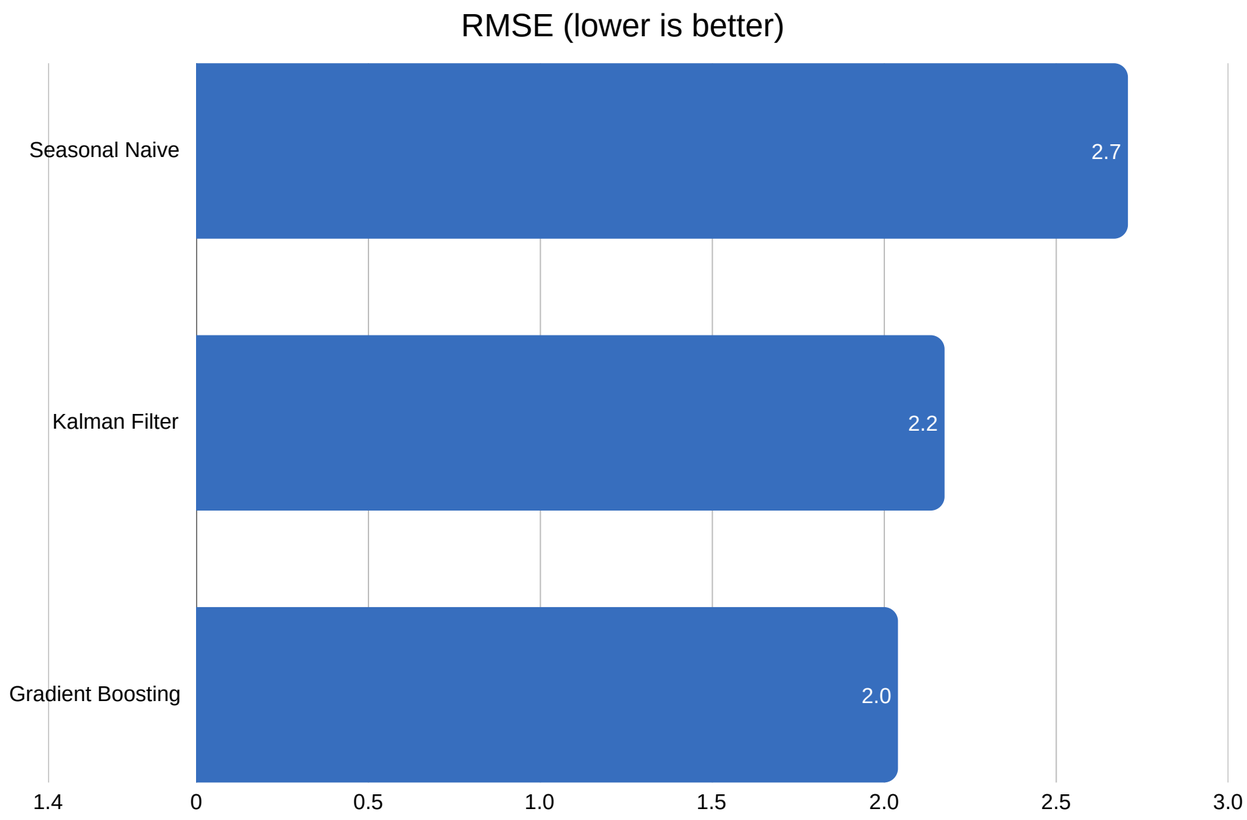
Role: main predictive model.

Evaluation metrics MAE = average absolute error • RMSE = penalizes larger errors • WAPE = absolute error relative to total demand

OVERALL EVALUATION

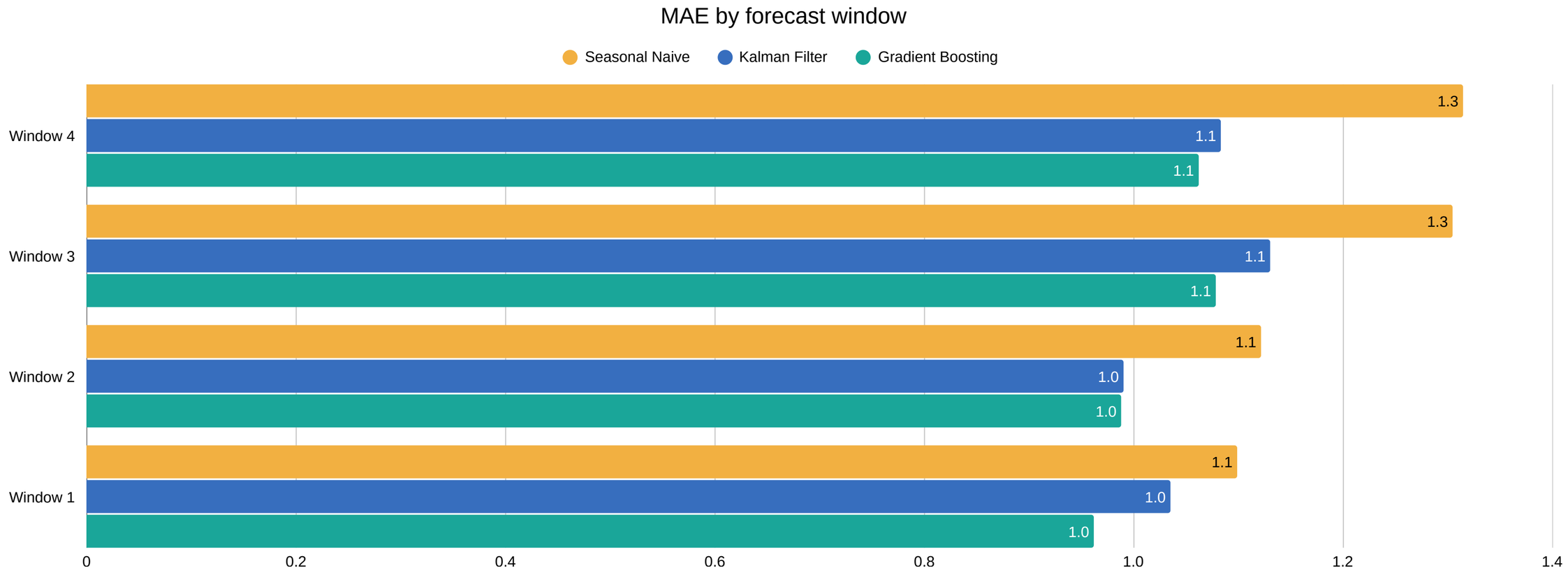
Results: Gradient Boosting leads





Gradient Boosting improves MAE by about 15.5% versus Seasonal Naive.

Performance is consistent across windows



Gradient Boosting has the lowest MAE in all four 7-day windows.

Uncertainty estimates from Kalman filtering

Quantiles

Approximate 10th, 50th, and 90th percentiles

Pinball loss

Mean pinball loss: 0.3413

Coverage

Observed 80% interval coverage: 87.57%

Interpretation

- The interval covered more actual values than the nominal 80% target in this sample.
- This suggests conservative/wide intervals for the evaluated slice not guaranteed calibration on all M5 series.
- The Kalman model is close to Gradient Boosting on point accuracy, while adding an uncertainty signal.

DISCUSSION

Limitations & next steps

Current limitations

- 100 sampled series, not the full M5 hierarchy
- Four rolling origins and fixed hyperparameters
- No explicit stockout labels
- Not the official M5 WRMSSE metric

Phase 2 opportunities

- Expand the sampled series and evaluation
- Tune hyperparameters on a separate validation period
- Test richer price and event signals where available
- Explore deep-learning models such as LSTM or DeepAR

Takeaway

A global Gradient Boosting model is the strongest point-forecasting approach in this controlled Phase 1 comparison; the Kalman filter contributes a useful uncertainty perspective.

KEY TAKEAWAYS

From historical sales to actionable forecasts

Best point accuracy

Gradient Boosting: MAE 1.0224

Reliable comparison

Same 2,800 points per model

Next step

Broader testing and tuning

Thank you